ENIGMA DARK

Securing the Shadows



Security Review

Euler

EVC #209 / EVK Periphery #445

Euler Earn #46

Nov, 2026

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Summary

Enigma Dark
Enigma Dark is a web3 security firm leveraging the best talent in the space to secure all kinds of blockchain protocols and decentralized apps. Our team comprises experts who have honed their skills at some of the best auditing companies in the industry. With a proven track record as highly skilled white-hats, they bring a wealth of experience and a deep understanding of the technology and the ecosystem.

Learn more about us at enigmadark.com

Euler EVC #209, evk-periphery #445 & euler-earn #46
ethereum-vault-connector PR #209 makes `EVCUtil` reject EVC authentication during an `isValidSignature` call, closing the permit context-spoofing issue (#630).

evk-periphery PR #445 rebuilds the governor as `GovernorAccessControlV2` on the patched EVC library and reissues the emergency governor and its factory as V2.

euler-earn PR #46 fixes the Permit2 fallback so a deposit still succeeds when the owner's ERC20 allowance to Permit2 is too low.

The three fixes are correct and complete for the code in scope. One informational note follows.

Engagement Overview

Over the course of 1 day, in November 2026, the Enigma Dark team conducted a fix review of the Euler EVC #209, evk-periphery #445 and euler-earn #46 changes. The review was performed by one Lead Security Researcher: vnmrtz.

The following repositories were reviewed at the specified commits:

Repository	Commit
ethereum-vault-connector	`diff e66a518e...883f048b` (PR #209)
evk-periphery	`diff 475b0514...7c5b7249` (PR #445)
euler-earn	`diff e42a2a45...7e8a1396` (PR #46)

Scope

The review was limited to the changes between the above commits, covering the following contracts:

```
ethereum-vault-connector/
└── src/utils/EVCUtil.sol

evk-periphery/src/
├── AccessControl/SelectorAccessControl.sol
├── Governor/
|   ├── GovernorAccessControlV2.sol
|   └── GovernorAccessControlEmergencyV2.sol
└── GovernorFactory/
    ├── GovernorAccessControlEmergencyFactoryV2.sol
    └── interfaces/IGovernorAccessControlEmergencyFactoryV2.sol

euler-earn/
└── src/libraries/SafeERC20Permit2Lib.sol
```

Risk Classification

Severity	Description
Critical	Vulnerabilities that lead to a loss of a significant portion of funds of the system.
High	Exploitable, causing loss or manipulation of assets or data.
Medium	Risk of future exploits that may or may not impact the smart contract execution.
Low	Minor code errors that may or may not impact the smart contract execution.
Informational	Non-critical observations or suggestions for improving code quality, readability, or best practices.

Vulnerability Summary

Severity	Count	Fixed	Acknowledged
Critical	0	0	0
High	0	0	0
Medium	0	0	0
Low	0	0	0
Informational	1	1	0

Findings

Index	Issue Title	Status
I-01	CER-18 states the permit deadline comparison backwards	Fixed

Detailed Findings

High Risk

No issues found.

Medium Risk

No issues found.

Low Risk

No issues found.

Informational

I-01 - CER-18 states the permit deadline comparison backwards

Severity: Informational

Context:

- docs/specs.md
- src/EthereumVaultConnector.sol#L520

Technical Details:
At the time of review, CER-18 stated the deadline condition as the `deadline` being not greater than `block.timestamp`. The code reverts when `deadline < block.timestamp`, so a valid permit requires `deadline >= block.timestamp`. The specification wording was inverted relative to the implementation, which is correct.

Recommendation:
State the condition as `deadline >= block.timestamp`.

Developer Response: Fixed at commit `7960920f`.

Disclaimer

This report does not endorse or critique any specific project or team. It does not assess the economic value or viability of any product or asset developed by parties engaging Enigma Dark for security assessments. We do not provide warranties regarding the bug-free nature of analyzed technology or make judgments on its business model, proprietors, or legal compliance.

This report is not intended for investment decisions or project participation guidance. Enigma Dark aims to improve code quality and mitigate risks associated with blockchain technology and cryptographic tokens through rigorous assessments.

Blockchain technology and cryptographic assets inherently involve significant risks. Each entity is responsible for conducting their own due diligence and maintaining security measures. Our assessments aim to reduce vulnerabilities but do not guarantee the security or functionality of the technologies analyzed.

This security engagement does not guarantee against a hack. It is a review of the codebase during a specific period of time. Enigma Dark makes no warranties regarding the security of the code and does not warrant that the code is free from defects. By deploying or using the code, the project and users of the contracts agree to use the code at their own risk. Any modifications to the code will require a new security review.